Flood Risk Assessment

APPLICATION REFERENCE	90.86425.FUL
SITE	17 Northgate Close, Balby, Doncaster, ZZ92 8CM

DOCUMENT DETAILS

Tomplan Planning and engineering Service

Architectural Drawing and engineering Service

Submitted to Council to accompany an application.

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www.tomplan.co.uk

Flood Risk Assessment (FRA)

Site: 55 St Catherine’s Avenue, Doncaster ZZ92 8CM

Proposal: Change of use from 1 C3 dwelling to 2 self-contained C3 flats with a modest

first-floor rear extension

Scope and Planning Context

This Flood Risk Assessment supports a Full Planning Application for redeveloping a residential

property into a two-flat unit with an additional rear extension.

The site is located on an established residential street, and the proposal does not introduce

a more vulnerable land use than the existing lawful residential use. The development

is therefore fully aligned with the principles of the National Planning Policy Framework
(NPPF)

and the Doncaster Strategic Flood Risk Assessment (SFRA).

The proposal satisfies the Sequential Test because it involves the reuse of existing
residential

land within the urban area and does not require application of the Exception Test.

The assessment demonstrates that flood risk has been appropriately considered, avoided

where possible, and mitigated through proportionate design measures, in accordance

with Doncaster Local Plan policy expectations.

FRA focuses on:

including the Environment Agency, and integrating it into a highmediumlow probability

and consequence matrix. This approach ensures clear planning by ranking flood risk sources,

including fluvial, surface water, groundwater, and reservoirs.

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Site Description

The site comprises an existing two-storey terraced dwelling within a built-up residential area of Doncaster. The surrounding context is entirely residential.

The proposed works are predominantly internal, involving subdivision into two self-contained flats, with a small rear extension at first-floor level. The scheme retains the established residential character and does not materially increase the ground-level building footprint beyond typical domestic curtilage works.

Flood Risk Sources Summary (Environment Agency Long-Term Flood Risk)

The Environment Agency's Long-Term Flood Risk information (downloaded on October 15, 2023) for 55 St Catherine's Avenue, ZZ92 8CM indicates the following:

Surface Water (Pluvial / Flash Flooding)

Surface water flooding is identified as the primary flood risk pathway affecting the site. Under

extreme rainfall events, shallow surface water accumulation is possible. For assessment purposes, envision this as 'ankle-deep water,' a peak runoff depth in the order of 150200 mm, considered a reasonable worst-case design assumption during intense storm events.

Rivers and the Sea (Fluvial / Tidal)

Groundwater

Reservoirs

Conclusion:

The dominant risk to the site is surface water flooding, particularly under future climate change scenarios. Fluvial flood risk is considered low, with an annual probability of less than

INTRODUCTION

1% (classified in the low probability band) now and under future projections.

For clarity, a "medium yearly chance" of surface water flooding may be interpreted as approximately a 510% annual probability during the 20402060 period, underscoring the importance of appropriate surface water management rather than site avoidance.

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Vulnerability and Flood Zone Considerations

The existing and proposed uses are residential (Use Class C3); therefore, the development does not increase flood vulnerability.

No basement or below-ground accommodation is proposed.

The scale of development is modest and, with appropriate drainage design, will not increase flood risk elsewhere.

While formal Flood Zone confirmation should be verified via the Environment Agency Flood Map for Planning, the available long-term risk data indicates low fluvial risk, consistent with a low-risk fluvial context.

Impact of the Development on Flood Risk Elsewhere

The development is designed to ensure that post-development runoff will be no worse than existing, and where feasible, improved. To verify and sustain the targeted 30% reduction in surface water runoff rates, a monitoring plan will be implemented. This plan will include bi-annual inspections of the drainage system and data collection on runoff rates, which will be compared against baseline pre-construction data to ensure compliance. Adjustments to the system will be made as necessary to achieve and uphold the targeted reduction.

As a design objective, the scheme will aim to: Achieve a minimum 30% reduction in surface water runoff rates, or Match greenfield equivalent runoff rates, where practical.

This measurable target converts policy intent into a clear, deliverable commitment, ensuring the development enhances rather than burdens local drainage infrastructure.

Key principles: Retain existing drainage routes. Control peak discharge during heavy rainfall events. Ensure the rear extension does not generate uncontrolled runoff from additional hardstanding.

Finished Floor Levels and Flood Resilience

Given the identified future surface water risk, the following resilience measures are proposed: Finished floor levels: Existing internal floor levels retained; no lowering proposed.

Thresholds: External door thresholds maintained or raised where feasible, typically 150 mm

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above external ground level, without compromising accessibility.

Materials: Use of moisture-resistant finishes at ground-floor level (e.g. tiled or vinyl flooring,

cement-based backing boards in wet areas). These measures reduce repair costs and improve

long-term asset resilience.

Services: Electrical consumer units and key services located at sensible heights above floor level

where practicable. These measures reduce damage potential and speed recovery following

extreme rainfall events.

Safe Access and Egress

Existing access from St Catherine’s Avenue is retained. Surface water flooding events are

anticipated to be short-duration and localised, ensuring that should flooding occur, egress

at ground level remains viable. The brief nature of such floods means that any evacuation can

be conducted safely and swiftly, minimizing risk to residents and reducing concerns over

accessibility during these events.

No basement accommodation is proposed.

Escape routes remain at ground level.

An evacuation awareness protocol will be implemented for residents, particularly

for vulnerable or mobility-impaired occupants, ensuring a timely response during

extreme events.

Final confirmation of local exceedance routes and street low points can be addressed

at the detailed design stage.

Summary

The Environment Agency’s long-term flood risk data confirms that the site’s primary flood risk

is surface water flooding, with low fluvial risk both now and under future climate change

scenarios. To ensure long-term stewardship, we commit to updating the Flood Risk Assessment

if new data or regulatory guidelines emerge. This forward-looking approach will help maintain

compliance with national and local flood risk policy while enhancing the development’s climate

resilience and adaptability.

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The proposed conversion to two self-contained flats, with a modest rear extension,

is acceptable in flood risk terms provided that:

Surface water runoff is controlled and reduced using proportionate SuDS measures.

Internal floor levels are maintained, and basic flood resilience is incorporated.

Exceedance routing safely directs water away from the building.

These measures provide a safe, sustainable, and policy-compliant development, while ensuring

that flood risk is not increased elsewhere and that long-term asset resilience is enhanced.

The proposal, therefore, satisfies national and local flood risk policy requirements and

is suitable for approval. Furthermore, aligning this modest scheme with Doncaster’s broader

resilience goals underscores its significance as a model for urban adaptation. This project
can

serve as a micro-scale demonstration of sustainable practices that contribute to the town’s

overarching resilience agenda, ultimately enhancing community preparedness

and environmental stewardship.